Excessive Harm*

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The horror! The horror!
Joseph Conrad (1899)

Denk ich an Deutschland in der Nacht,
Dann bin ich um den Schlaf gebracht.
Heinrich Heine (1844)

Abstract

A father punishes a son who has done a misdeed; a mother can sanction the father to protect the son. Both sides harbour private information: the father may be just or harsh, the mother's resolve to carry out sanctions strong or weak. I characterise when excessive harm is inflicted and how it depends on the mother's sanctioning costs and the father's zeal. Invariably, protection runs through early sanctions—feared or imposed—even when the father has not yet crossed the line. The model speaks to international law, where punisher and protector are states.

1 Introduction

A son has conducted a misdeed. A father wants to punish—perhaps justly, perhaps far beyond. A mother stands by watching how punishments unfold. She has one instrument though: she can sanction the father after each round of punishments.

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I shall stick to this allegory throughout, although the applications I am interested in are much broader, including an application to international law where a state actor is represented by the father, a population that bears the punishment by the son, and third states by the mother.

The father has two types—a just father who wants to punish justly and a harsh father who wants to punish excessively. The mother also has two types, which vary in the costs of sanctioning: a strong mother with lower costs and a weak mother with higher costs. The mother's main objective is to protect her son from excessive punishment, and the paper's main objective is to characterise the situations where she fails in this endeavour.

There is a trivial failure and a not-so-trivial one. The trivial failure arises when sanctioning is too dear for any threat to be worth making: even a strong mother then stays silent. The non-trivial failure arises from the second source of incomplete information, the mother's resolve, which may be tested by the father's first strike, screening the mothers. There are parameter bands where a strong mother can shelter her son but a weak mother cannot—because the pooling equilibria that would protect all sons break down. The father's zeal will play a role here, too.

It is worth saying at the outset where the paper's weight lies. The first failure is a familiar tragedy: when threats are too costly, silence will follow and the father has free rein. The second is the one this paper exists for. Proof of resolve is possible here—a resolute mother can signal her type—but precisely because signalling is possible, a weak mother cannot hide and the father may screen. In that case the very technology that protects one son unmasks the other's mother. Uncertainty, while it lasts, may be a shelter for the weak mother's son, and I will show when it breaks down and how the son will suffer then.

This paper was written with Gaza in view. It would be coy to pretend otherwise. But the reader should know exactly what is, and is not, being claimed. Father, mother, and son are mnemonic labels, chosen for compression, but they apply to any similar situation in which there is a punisher, a passive entity to be punished, and a third party that can intervene through sanctions. The model does not adjudicate the conflict. It takes as given that some level of punishment is just and that the harsh punisher's ideal exceeds it. It deliberately gives the son no

moves—an abstraction isolating the punishment–sanction relation.

How time unfolds is an important determinant of that relation. In order to simplify as much as possible without losing the gist, I study a model where punishment unfolds over two periods. After the first punishment the mother can immediately impose a sanction on the father and she can announce a sanctioning scheme for excessive punishments in round 2. Whether these later sanctions are carried out depends on the mother's sanctioning costs.

This two-period structure allows for the analysis of the interplay between the mother's uncertainty about the father's type and the father's uncertainty about the mother's type. Weak mothers will have an incentive to pool with strong mothers, but harsh fathers may, as it turns out, prefer to screen the mothers and reveal their type by doing so. If that screening succeeds—because the weak mother cannot afford to pool with the strong mother—the non-trivial path to excessive punishment arises.

The paper proceeds as follows. Section 2 places the model in the literature. Section 3 sets out the model, and Section 4 solves the game through backward induction up to stage 2. Section 5 then examines the father's first strike and states the main theorem, characterising situations where excessive punishments unfold. Section 6 pauses on zeal, the model's most malign parameter and its subtlest. Section 7 discusses the results and Section 8 offers a coda.

2 Related literature

This section maps the paper's neighbourhoods. It has four main ones.

Signalling, and the price of being believed. The paper's machinery descends from Spence (1973): a costly action separates types when its cost differs across them. Riley (1979) characterised the least-cost separating outcome our equilibrium selects, and the refinement that disciplines beliefs off the path is the Intuitive Criterion of Cho and Kreps (1987), with Banks and Sobel (1987) the neighbouring divinity refinement. Two features place the present model in a particular corner of that tradition. First, the period-one sanction is, in the language of the literature, *burned money*: an action whose main function is to be seen to have been costly. Ben-Porath and Dekel (1992) showed how the ability to burn money before play selects outcomes; Austen-Smith and Banks (2000) married burning to cheap talk

and established when the match adds credibility. A mother's initial sanction is just that—burned money, with one additional feature: it harms the receiver. The immediate sanction is free to the resolute sender, dear to the pretender, and a direct penalty to the father—which is why it enters his calculations before the first strike, and not only his beliefs. The mother's announcement of a sanctioning scheme for further potentially excessive punishments could be modelled as cheap talk in the tradition of Crawford and Sobel (1982) but we will depart from that by assuming posturing costs in the spirit of costly-talk models of Kartik et al. (2007) and Kartik (2009).

Resolve, and how to demonstrate it. That threats are instruments, that their credibility is the question, and that commitment is a strategic act are the inheritance of Schelling (1960, 1966)—who, it is worth recalling, reached repeatedly for the disciplining of children as his illustration of choice: the family frame has a pedigree. The formal study of resolve descends through the reputation models of Kreps and Wilson (1982) and Milgrom and Roberts (1982). The natural habitat of these ideas is interstate conflict, where war among rational actors is itself an equilibrium phenomenon (Fearon, 1995; Powell, 2006)—and where Fearon (1997) drew the distinction our instrument pair embodies: *sinking costs* versus *tying hands*. The period-one sanction is sunk; the announced rate ties no hands at all—it is the limiting case Fearon's dichotomy leaves implicit, speech whose only collateral is the posturing paid to utter it. Audience-cost and reputational theories of communication (Fearon, 1994; Schultz, 1998; Sartori, 2002) supply what our announcements conspicuously lack, an external penalty for bluffing; the model shows how much sorting survives without one. Demonstrating resolve on behalf of someone else is the business of extended deterrence—Fuhrmann and Sechser (2014) contrast hand-tying and sunk costs in exactly that setting. What is different here is where the signal lands: it runs through sanctions on the punisher himself, in the middle of the punishment, and he can choose to screen it.

Sanctions. The economics of sanctions has a long tradition with Kaempfer and Lowenberg (1988) and Eaton and Engers (1992) being a good starting point, the latter the canonical model of a sender ensuring compliance through threatened harm, compressed in Eaton and Engers (1999). Tsebelis (1990) stated the game-theoretic paradox that tighter sanctions need not improve behaviour. The selection insight—that observed sanctions are already failures, because threats that would

work, work as implicit threats—runs through Smith (1995), Drezner (2003), and Hovi et al. (2005), with Drezner (1999) the book-length statement and Lacy and Niou (2004) the issue-linkage variant; Morgan and Schwebach (1997) and Bapat and Kwon (2015) give the empirical and enforcement-theoretic counterparts. Our bands of protection failure are cousins of that selection logic with the mechanism relocated: threats are absent not because they would succeed unspoken but because announcing them costs more than they would spare. Closest in spirit is Verdier (2009), for whom sanctions are revelation regimes—though the direction here is reversed: there the sanctioner designs the game that reveals the target's type; here the target of sanctions, the father, may choose to screen.

Punishment, and punishing the punisher. Finally, the model inverts a familiar architecture. In the economics of punishment from Becker (1968) onward, and in the experimental literature on costly and third-party punishment (Fehr and Gächter, 2000; Fehr and Fischbacher, 2004), sanctions discipline a wrongdoer on behalf of a community. Here the object of discipline is the punisher himself: the mother is a third-party punisher of a punisher, and the question is whether such second-order enforcement can protect the first victim. The answer the model returns—that it protects him only through the mother's resolve for further punishments being provable—rests on a combination that is, to my knowledge, new: an endogenous first strike that sets the stakes, type revelation through a signal that penalises the punisher directly, and the bill delivered to a passive third party.

3 The model

A father plans to punish a son who has conducted a misdeed. His ideal punishment depends on his type, which only he knows: a just father wants a just amount, a harsh father an excessive amount. A mother cannot stop him directly, but she can make him pay. She can impose an immediate penalty on him after his first strike, and she can threaten a further penalty to come after his second. Whether she would actually follow through on the threat depends on her own resolve, which only she knows.

The order of events is as follows. The son has conducted his misdeed. That is in the past and irreversible. It is not part of the game I study. Rather, the game begins with the father's first strike. The mother, having observed the inflicted

punishment, responds: she imposes a penalty now and announces a penalty scheme for what comes later. The father, having observed her response, strikes a second time and total punishment is then realised, as we restrict our analysis to two periods. Next, the mother decides whether to carry out the threatened penalty or not.

This is a game with two-sided incomplete information: initially, the mother does not know the father's type, nor does the father the mother's. But as the game unfolds, both will update, and in some cases the updating will collapse to certainty.

Figure 1 summarises the game's temporal structure.

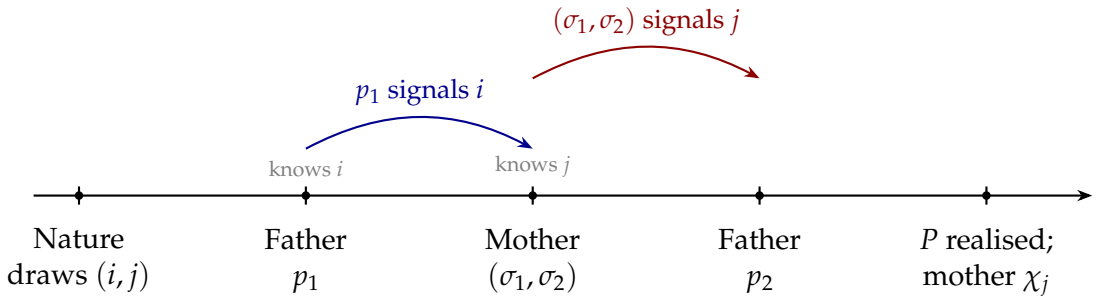


Figure 1: The order of play and the two signalling channels: the father's period-one punishment may signal his type to the mother, and the mother's response may signal her type to the father. At the final node $\chi_j \in \{0, 1\}$ is the mother's carry-out decision.

Players and types. There is a father F, a mother M, and a son. The son is passive: he takes no action and is the object of the punishment; only F and M choose. Nature draws the two types independently,

$$i \in \{J, H\}, \quad \Pr(i = H) = \pi \in (0, 1), \qquad j \in \{S, W\}, \quad \Pr(j = S) = \rho \in (0, 1),$$

where i indexes the father's type and j the mother's. Each player observes only its own type.

Punishment. Punishment is delivered over two periods. In period $\tau \in \{1, 2\}$ the father inflicts $p_\tau \in [0, \bar{p}]$, where $\bar{p}$ is the per-period capacity; total punishment is $P = p_1 + p_2$. The father's ideal total is $y_J = 1$ for the just type and $y_H = \theta > 1$ for the harsh type. The mother's ideal mimics the just father's ideal. She, too, wants just punishment to be carried out.

There is no disagreement among the players about what a just punishment entails, which allows us to make the normalisation to 1.

The mother's instruments. After observing her own type and p_1, the mother picks two numbers: a period-one sanction σ_1 imposed immediately, and an *announced rate* σ_2 for period two, with $\sigma_1 \geq 0$ and $\sigma_2 \geq 0$: neither instrument is bounded above. The rate scheme threatens the father with a sanction $\sigma_2 (P-1)$ proportional to the excess of the son's punishment over the just level, imposed if the father pushes past $P = 1$ and the mother carries the scheme out. As a strategy these are rules $\sigma_1(p_1; j)$ and $\sigma_2(p_1; j)$ mapping the observed p_1 into the two numbers; at the point of choice they are simply scalars. At the final stage, after P is realised, the mother chooses whether to execute the announced scheme, $\chi_j \in \{0, 1\}$.

Sanctions and costs. The realised sanctions are

$$\Sigma_1 = \sigma_1, \qquad \Sigma_2 = \chi_j \, \sigma_2 \, (P-1) \, \mathbf{1}\{P > 1\} :$$

the period-one sanction is imposed in full by both types, while the period-two sanction, proportional to the excess $P - 1$, is realised only if the mother carries it out at the end. She bears two linear costs: (i) a *posturing cost* $\gamma\sigma_2$—the apparatus, preparation, and political capital that announcing a sanctions regime consumes—with $\gamma > 0$, per unit of the *announced* rate σ_2, is paid by both types in period one upon announcing, whether or not the scheme is ever executed; it attaches only to the announcement, not to the act of carrying out. (ii) a *type-dependent cost* at rate c_j, with $c_S = 0$ and $c_W > 0$, that applies to any sanction *actually imposed*—to Σ_1 and, when carried out, to Σ_2. The zero for the strong type is a net figure, not a claim that sanctioning is free: her costs of imposing sanctions are as real as the weak type's, but they are offset by an intrinsic return—she values upholding an order in which excess is answered, and that stake pays the bill. The weak type holds no such stake, so for her the bill stands; c_j is the net cost of keeping one's word.

It is worth noting that, at the final stage, the strong mother will behave just like a commitment type would: she executes what she announced, because nothing stops her—carrying out the sanction is costless and a convention of resolve introduced below closes the residual indifference. But she is not merely a commitment type—her advantage is already present in period one, where the immediate sanction is free to her and costly for the weak type, and it is there—

before any excess has occurred—that resolve can be proven. Section 7 shows that a model in which resolve manifests only at the moment of execution delivers none of the sorting the remainder of the paper builds on.

Payoffs. The father of type i receives

$$U_i = -\beta(P - y_i)^2 - \Sigma_1 - \Sigma_2, \qquad \beta > 0, \quad i \in \{J, H\},$$

where the scalar $\beta > 0$ is the intensity of the punishment motive: it weights the deviation from the ideal y_i against the sanctions.

The mother of type j receives

$$U_j = -\ell(P) - c_j\Sigma_1 - \gamma\sigma_2 - c_j\Sigma_2, \qquad j \in \{S, W\},$$

with the quadratic loss

$$\ell(P) = (P - 1)^2.$$

Written out by type, using $c_S = 0$ and $c_W > 0$, this is

$$U_S = -\ell(P) - \gamma\sigma_2, \qquad U_W = -\ell(P) - \gamma\sigma_2 - c_W\big(\Sigma_1 + \Sigma_2\big) :$$

the strong mother pays only for posturing; the weak mother pays, on top, for every sanction she actually imposes.

Strategies and beliefs. The father's period-one strategy is a number $p_1(i) \in [0, \bar{p}]$ for each type; his period-two strategy is a map $p_2(i, p_1, \sigma_1, \sigma_2) \in [0, \bar{p}]$, a function of his type, his own period-one choice, and the observed sanction pair.

The mother's strategy is the pair of rules $\sigma_1(p_1; j), \sigma_2(p_1; j)$ together with the carry-out χ_j. Strictly, this should also be a mapping from the game's full history onto a binary decision but, as we will see, our assumptions about costs (and an additional tie-breaker assumption below) allow us to conflate it, considerably simplifying notation.

Two posteriors, one for each signalling channel, complete the description. After observing p_1, the mother updates on the father's type,

$$\hat{\pi}(p_1) = \Pr\big(i = H \mid p_1\big),$$

read against the father's period-one strategy $p_1(\cdot)$; before choosing p_2, the father updates on the mother's type,

$$\hat{\rho}(\sigma_1, \sigma_2; p_1) = \Pr(j = S \mid \sigma_1, \sigma_2, p_1),$$

read against the mother's rules at the realised p_1.

Two restrictions on parameters are maintained throughout.

Assumption 1. $\frac{1}{2} < \bar{p} < 1$ and $\theta \leq 2\bar{p}$.

Assumption 2. $\beta\gamma < \theta - 1$.

Assumption 1 ensures that neither ideal can be attained in a single period and that both ideals are attainable in two. Assumption 2 gives deterrence a chance at all: it will emerge below that when $\beta\gamma \geq \theta - 1$, no mother ever finds any sanction worth its posturing, and the analysis collapses to the trivial regime in which a harsh father simply takes his ideal.

4 Backward induction

This section solves the game from the back—the carry-out decision, the second strike, the mother's response to the father's first—and stops one step short of the beginning: the very first strike from the father, which warrants its own section further below (Section 5).

We solve for the game's perfect Bayesian equilibria: a profile of strategies for the father and the mother together with the posteriors $\hat{\pi}(p_1)$ and $\hat{\rho}(\sigma_1, \sigma_2; p_1)$, such that every action is sequentially rational given beliefs and continuation play, and beliefs are derived from strategies by Bayes' rule. We restrict attention to pure strategies, and we require equilibria to survive the Intuitive Criterion of Cho and Kreps (1987), applied to the mother's signalling problem at each realised p_1.

A general analysis of the game would leave behaviour undetermined at several points, a common issue with games of incomplete information—due to ties, knife edges, and multiplicity of beliefs—and rather than resolve them piecemeal, we gather every selection rule into one place and use it consistently. All results below are statements about the equilibrium this block selects—a characterisation of the selected equilibrium, not of the full set of perfect Bayesian equilibria.

Definition 1 (The selected equilibrium)**.** Among the pure-strategy perfect Bayesian equilibria surviving the Intuitive Criterion, we select lexicographically by three conventions.

(R1) *Resolve.* The strong mother is lexicographically resolute: whenever she is otherwise indifferent at the final stage and the threatened sanction is positive, she carries out the announced sanction. This is where credibility enters the model, and it enters as a convention of resolve, not as a derived incentive (Remark 2).

(R2) *Parsimony.* No player takes a costly or idle action that serves no purpose: no mother imposes a sanction that neither disciplines nor informs; no father arms beyond what his ideal requires; the just father opens at the least reach that permits $P = 1$; and among signalling configurations that leave the sender's continuation payoff unchanged, the least signal is selected. In particular, immediate sanctions above the least separating level are discarded, and a weak mother exactly indifferent between revealing and mimicking the strong type's costly signal reveals.

(R3) *Quiet-side boundary defaults.* Knife-edge thresholds are closed on the quiet side: a strong mother exactly indifferent at the reticence-band edge—where the harm the threat would spare just equals its posturing cost—announces nothing, and a harsh father exactly indifferent between parking at the band edge and striking into separation parks. Given the just father's least opening, reaches above the just level are read as harsh: $\hat{\pi}(x) = 1$ for all $x > 1$.

Remark 1 (What the selection does)*.* One convention is substantive on its face: (R1), resolve, which tilts the selection toward enforcement. The others resolve ties—and ties, here, sit on points the players seek out, so resolving them is outcome-relevant selection, not housekeeping. Parsimony strips waste—idle and over-sized sanctions, arms beyond the ideal—places the just father at his least sufficient opening, and resolves the weak mother's tie at mimicry indifference toward revelation. That last resolution leans against enforcement in form only: a weak type playing the strong bundle turns the configuration into a pool, Bayes' rule at the bundle diluting the very credence the mimicry was to exploit, and every such pool falls to the refinement (Lemma 9 below); the pretence, played, unravels what it borrows. The boundary defaults close knife edges of measure zero on the

quiet side—the reticence band keeps its upper edge, parking is selected at first-strike indifference—which is what lets the father's problem attain its optimum: he will be seen to probe the very boundary of what the mother tolerates, and that boundary must be his to stand on. And the belief $\hat{\pi}(x) = 1$ extends the least opening: with the just father at reach 1, every reach above the just level is vacated for the harsh type alone. This is the belief most favourable to deterrence—just-father pooling at higher reaches would inflate the target excess $b = \beta\gamma/\hat{\pi}$ and, at any fixed reach, *widen* the band of unanswered aggression. One convention, however, is not innocent, and it should be read in the open: the least sufficient sanction. Over-sized separating sanctions survive the refinement (Lemma 8), and a convention of over-proving would raise the father's cost of entering separation—enough, on an open set of primitives, to tip him into parking and to *lower* the son's expected punishment, at no cost to the strong mother, who is indifferent to the size of a burn she does not feel. Proof in excess of necessity is itself a deterrent; the least-signal convention declines it. There is no deeper defence: nothing in the strong mother's incentives pins the size of a burn she does not feel, so the least signal is the literature's convention, not the model's prediction—and this remark prices what it discards. The theorems of this paper are therefore statements about the standard, least-cost selection—not about enforcement at its best.

4.1 The final stage

We begin at the last node, where the mother decides whether to carry out the announced sanction.

Lemma 1. *At any final history with a positive threatened sanction, $\sigma_2(P-1)\,\mathbf{1}\{P > 1\} > 0$, the weak mother strictly declines to carry it out and the strong mother is indifferent. At histories where the threatened sanction is zero, the carry-out decision is vacuous and parsimony selects no execution for both types.*

Proof. Immediate: all other terms are sunk, and executing costs the weak type $c_W\sigma_2(P-1) > 0$ and the strong type nothing, since $c_S = 0$. A zero threatened sanction makes χ_j payoff-irrelevant for both types, an idle action removed by (R2). □

Remark 2 (The knife edge of resolve). By (R1) the strong mother carries out whenever there is anything to carry out. Her indifference is exact, since $c_S = 0$, and

deserves a word rather than concealment. One reading is definitional: the strong type *is* the type for whom keeping her word costs nothing—resolve modelled as the absence of any reason to flinch. The other is a perturbation: an arbitrarily small expressive or reputational payoff to execution for the strong type breaks the indifference in favour of carrying out; (R1) is the vanishing limit of that perturbation. The same parameter makes the period-one sanction costless for her, so the benchmark throughout is one in which *proof is costly only for the wrong type*—the cleanest case for the signalling logic. An expressive or reputational benefit to execution, exceeding a small positive c_S, gives the strong type a strict final-stage incentive to carry out and so replaces (R1) with an optimum. It also enters her earlier choices—the benefit scales with what is executed, so the optimal announcement, and with it the least separating sanction, shift. Working the perturbed model out is left for later work; the benchmark throughout is $c_S = 0$ with (R1).

4.2 The father's second strike

The just father. At the fourth stage the father chooses $p_2 \in [0, \bar{p}]$, adding to the sunk p_1, so the reachable totals form the interval $[p_1,\ p_1 + \bar{p}]$; since $\bar{p} < 1$ we have $p_1 < 1$ throughout. The just father wants $P = 1$, and as $p_1 < 1$ the only question is whether he has the capacity to reach it.

Lemma 2. *The just father sets $p_2 = 1 - p_1$ if $p_1 + \bar{p} \geq 1$, attaining $P = 1$; otherwise he sets $p_2 = \bar{p}$, attaining $P = p_1 + \bar{p} < 1$. In either case $P \leq 1$, so he never enters the sanctioned region, and the period-two sanction is irrelevant to his choice.*

Proof. His loss $\beta(P-1)^2$ is increasing in the distance of P from 1, and the reachable set is $[p_1, p_1 + \bar{p}]$ with $p_1 < 1$. The closest point to 1 is 1 itself when $1 \leq p_1 + \bar{p}$, and $p_1 + \bar{p}$ otherwise. Neither exceeds 1, so the sanction term never binds. □

The harsh father. The harsh father wants $P = \theta > 1$, so unlike the just father he wishes to push past the just level. What restrains him is the *expected* marginal sanction he faces above $P = 1$. He has observed the announced rate σ_2, but it bites only if the mother carries the scheme out; assigning probability $\hat{\rho}$ to her being strong, he expects a marginal sanction $\hat{\rho}\sigma_2$ on each unit of excess. He therefore chooses P to maximise $-\beta(P-\theta)^2 - \hat{\rho}\sigma_2(P-1)\,\mathbf{1}\{P > 1\}$ over the reachable interval, whose top is

$$\bar{P} = \min\{p_1 + \bar{p},\ \theta\}.$$

The objective is concave, with unconstrained maximiser

$$P^{\circ} = \theta - \frac{\hat{\rho}\sigma_2}{2\beta}.$$

Write $r^{\star} = 2\beta(\theta - 1)$ for the *fully-deterring rate*: the expected marginal sanction at which the father is pulled all the way back to the just level.

Lemma 3. *The harsh father's second strike attains*

$$P^{\star} = \min\left\{\bar{P},\ \max\left\{1,\ \theta - \tfrac{\hat{\rho}\sigma_2}{2\beta}\right\}\right\}.$$

Equivalently, when $\bar{P} > 1$: he is held at $P = 1$ when $\hat{\rho}\sigma_2 \geq r^{\star}$ (full deterrence); he settles at the interior $P^{\circ} = \theta - \hat{\rho}\sigma_2/2\beta$ when $0 < \hat{\rho}\sigma_2 < r^{\star}$ and $P^{\circ} \leq \bar{P}$ (partial deterrence); and he is capped at the ceiling $\bar{P}$ otherwise, reaching his ideal θ when $p_1 + \bar{p} \geq \theta$. When $\bar{P} \leq 1$, the sanctioned region is unreachable and he chooses $P = \bar{P}$.

Proof. If $\bar{P} \leq 1$, the sanctioned region is unreachable, the scheme never bites, and the harsh father takes the largest feasible total, $P = \bar{P}$; the displayed formula returns exactly this, the inner maximum being at least 1. Let then $\bar{P} > 1$. Among totals below 1 no sanction is paid and the objective is strictly increasing toward $\theta > 1$, so any optimum lies in $[1, \bar{P}]$. On this interval the objective is concave with unconstrained maximiser P°, found from the first-order condition $2\beta(\theta - P) = \hat{\rho}\sigma_2$. Projecting P° onto $[1, \bar{P}]$ gives the stated $P^{\star}$: it equals 1 when $P^{\circ} \leq 1$, i.e. $\hat{\rho}\sigma_2 \geq 2\beta(\theta - 1) = r^{\star}$; it equals P° when P° lies inside; and it equals $\bar{P}$ when $P^{\circ} \geq \bar{P}$, the sanction then slack against the binding reach. Whenever $\bar{P} > 1$ the father never chooses below 1, which he prefers to any lower punishment; when $\bar{P} \leq 1$, capacity forces $P = \bar{P}$. □

Corollary 1. *To deter fully the announced rate must satisfy $\hat{\rho}\sigma_2 \geq r^{\star}$, that is $\sigma_2 \geq r^{\star}/\hat{\rho}$. The father's inference fixes the discount: a mother known weak ($\hat{\rho} = 0$) is never believed and never deters; a mother known strong ($\hat{\rho} = 1$) deters at the rate $r^{\star}$; under pooling ($\hat{\rho} = \rho$) the announced rate must be inflated to $r^{\star}/\rho$, since only the strong type would carry it out.*

4.3 The mother's problem at a given reach

In the third stage the mother makes two choices: the sanction σ_1 imposed directly, and the rate σ_2 she announces for period two, carried out if $P > 1$. One case can

be set aside at once.

Lemma 4. *If $p_1 + \bar{p} \leq 1$, no deterrence is possible: even a harsh father cannot reach past the just level, so both types of mother set $\sigma_1 = \sigma_2 = 0$, and the son's punishment is at most $p_1 + \bar{p}$ regardless of either type.*

Proof. The reachable totals form $[p_1, p_1 + \bar{p}]$ with top at or below 1, so $P > 1$ is impossible and the scheme can never be triggered. No sanction affects the outcome; by parsimony neither type imposes one. The father, of either type, goes to the top $p_1 + \bar{p}$. □

We therefore assume $p_1 + \bar{p} > 1$ throughout the remainder. It is convenient to carry the analysis in terms of the father's *reach*

$$x \equiv p_1 + \bar{p},$$

the largest total he can still attain; since striking to reach past θ changes nothing—a harsh father never goes beyond his ideal—we may restrict attention to $x \in (1, \theta]$ without loss. Throughout this subsection the reach x and the mother's posterior $\hat{\pi} = \hat{\pi}(p_1)$ are fixed; both are endogenised in Section 5.

Notation. Write $a = \theta - 1$ for the harsh father's excess ideal. When the father attaches credibility $\hat{\rho}$ to the scheme, a mother weighing harm against posturing faces, per unit of announced rate, a pull-back of $\hat{\rho}/2\beta$ on the father and a cost of γ; equating the marginal harm reduction $\hat{\pi}\hat{\rho}(P-1)/\beta$ to γ gives the *target excess*

$$b(\hat{\rho}) = \frac{\beta\gamma}{\hat{\pi}\hat{\rho}},$$

the distance above the just level at which deterrence is no longer worth pushing. Two values of the credibility will matter: full credibility, with $b \equiv b(1) = \beta\gamma/\hat{\pi}$, and pooled credibility, with $b_p \equiv b(\rho) = \beta\gamma/(\hat{\pi}\rho)$. Finally, for $z \in (0, a]$ define

$$\Delta(z) = \sqrt{z\,(2a - z)},$$

an increasing function with $\Delta(z) > z$ on $(0, a)$ and $\Delta(a) = a$. Its role is the following: $1 + z$ is where a mother *pulls the father to* if she deters at credibility-adjusted target excess z, while $1 + \Delta(z)$ is the reach below which she *declines to*

deter at all. The wedge between the two—the target is nearer the just level than the threshold for bothering—will do much of the work below.

We consider the two instruments in turn, asking through which the mother's types can separate—for it is separation that exposes the weak mother and so shapes the son's fate.

Remark 3. The sanction σ_1 is costly—it lowers the father's payoff by σ_1 and the mother's by $c_j\sigma_1$—yet it is sunk before the father's second strike, so it can bear on what follows only through the father's posterior. A positive σ_1 is therefore incurred only to signal—burned money, in the language of Section 2—and since it is free to the strong type but costs the weak type $c_W\sigma_1 > 0$, it is well suited to separating the types.

Lemma 5. *The types cannot separate through σ_2 alone.*

Proof. An announced rate σ_2 pulls the harsh father back continuously, by $\hat{\rho}\sigma_2/2\beta$, and costs the announcing mother the posturing $\gamma\sigma_2$; both the effect and the cost are the same whichever type announces it. So whatever rate the strong type announces, the weak type obtains the identical effect at the identical cost by matching it—and saving her son is worth that cost to her whenever it is worth it to the strong type. The rate σ_2 thus carries no differential that could make the types' choices diverge: any rate the strong type might announce to separate is one the weak type strictly wishes to match. One boundary point is the exception that proves the rule: at $x = 1 + \Delta(b)$ the value of being believed is exactly zero, both types are indifferent, and adverse beliefs can prop up a knife-edge separation through the announcement alone. The quiet-side default (R3) discards it; strictly above the band the strict preference returns. There is one more tie, at over-sized rates where the posturing eats up the whole value of being believed; adverse beliefs could prop up a knife-edge separation there too. The immediate sanction removes it: a slightly cheaper announcement backed by a small σ_1 is a deviation only the strong type can want, so the Intuitive Criterion forces the favourable belief and the knife edge collapses. Without that instrument the tie survives—a point Section 7 will need. □

Lemma 6. *The weak mother can reduce her son's punishment only by pooling with the strong type on a positive rate $\sigma_2 > 0$.*

Proof. Whenever the weak mother's type is revealed, the father learns she will not carry the scheme out, so he faces expected rate $\hat{\rho}\sigma_2 = 0$ and drives the son to the reach x, regardless of any rate she has announced. Only if she pools with the strong type does the father retain a positive belief $\hat{\rho} = \rho$; then an announced rate $\sigma_2 > 0$ leaves him facing expected rate $\rho\sigma_2 > 0$ and pulls him back below x. A pool on $\sigma_2 = 0$ secures no pull-back, so only pooling on a positive rate helps her. □

4.4 Reticence and deterrence

The strong mother believed. We look first at the benchmark in which the strong mother's type has been revealed, so the father holds $\hat{\rho} = 1$ and the weak type drops out of his reckoning. She still does not know the father's type, facing a harsh one with probability $\hat{\pi}$, and only a harsh father ever triggers the scheme. Her problem is to choose the announced rate σ_2 to minimise $\hat{\pi}\big(P(\sigma_2) - 1\big)^2 + \gamma\sigma_2$, where by Lemma 3 a harsh father responds with $P(\sigma_2) = \min\{x,\ \max\{1,\ \theta - \sigma_2/2\beta\}\}$.

Lemma 7 (Reticence and deterrence)**.** *Let $\hat{\rho} = 1$ and $b = \beta\gamma/\hat{\pi} < a$. The strong mother's announced rate takes one of two values:*

(i) *if $x \leq 1 + \Delta(b)$, she announces nothing, and a harsh father takes the reach x;*

(ii) *if $x > 1 + \Delta(b)$, she announces*

$$\sigma_2^\star \;=\; 2\beta\,(a - b),$$

holding a harsh father at $P = 1 + b$.

If $b \geq a$ she announces nothing at any reach. We call the interval $\big(1 + b,\ 1 + \Delta(b)\big]$ the reticence band*: reaches at which she could pull the father back to her target, yet declines, because the harm spared does not cover the posturing.*

Proof. Rates below $2\beta(\theta - x)$ leave the father at the reach—the scheme is announced but never bites—so on $[0, 2\beta(\theta - x)]$ the objective is $\hat{\pi}(x - 1)^2 + \gamma\sigma_2$, increasing, and the best point of the inactive range is $\sigma_2 = 0$. Rates $\sigma_2 \geq 2\beta a$ fully deter, all inducing $P = 1$, and are dominated by the lowest of them: same harm, higher posturing. On the active range $\big(2\beta(\theta - x),\ 2\beta a\big]$ the objective is

$\hat{\pi}(\theta - 1 - \sigma_2/2\beta)^2 + \gamma\sigma_2$, convex, with first-order condition $\hat{\pi}(P-1)/\beta = \gamma$, i.e. $P = 1 + b$, attained at $\sigma_2^\star = 2\beta(a-b)$; this lies in the active range precisely when $1 + b < x$. (Feasibility of the target: the required second strike is $p_2 = 1 + b - p_1$; from $1 + b < x = p_1 + \bar{p}$ it satisfies $p_2 < \bar{p}$, and $p_2 \geq 0$, i.e. $1 + b \geq p_1$, is equivalent to $x - 1 - b \leq \bar{p}$, which follows from $x \leq \theta = 1 + a$, $b > 0$, and $a \leq 2\bar{p} - 1 < \bar{p}$.) The choice is therefore binary: announce nothing, for $\hat{\pi}(x-1)^2$, or announce $\sigma_2^\star$, for $\hat{\pi}b^2 + \gamma\sigma_2^\star = \hat{\pi}b^2 + 2\hat{\pi}b(a-b) = \hat{\pi}\,b(2a-b) = \hat{\pi}\,\Delta(b)^2$. The strong mother thus weakly prefers deterrence iff $(x-1)^2 \geq \Delta(b)^2$, i.e. $x \geq 1 + \Delta(b)$; at equality, by (R3), the selected action is nevertheless silence ($\sigma_2 = 0$). Hence the selected action is silence for $x \leq 1 + \Delta(b)$ and deterrence strictly above. If $b \geq a$ the active-range objective is increasing throughout and she announces nothing. □

Two features deserve emphasis. First, when she deters, she stops short: even a strong mother, certain of a harsh father and believed without doubt, leaves her son at $1 + b$, a residual harm above the just level that shrinks only as posturing becomes free. Second, *before* she deters there is a band of reaches in which she does nothing at all. Within the reticence band the father faces no sanction of any kind, from any mother, however she is believed. The quiet region grows with $\beta\gamma$—its edge $1 + \Delta(b)$ climbs to θ as $b \to a$—though the band proper need not: its width $\Delta(b) - b$ rises only until $b = a(1 - 1/\sqrt{2})$ and then shrinks, the target chasing the edge and catching it at θ. Reticence swallows deterrence continuously as threatening grows dear.

A word on the band's edge, which is where the quiet-side default (R3) does its only work. Strictly inside the band the strong mother declines to deter as a strict inequality—the harm spared falls short of the posturing, and her low cost of *carrying out* is beside the point, since what does not pay is the *announcement*. Strictly above it she deters, again strictly. Only at the single reach $x = 1 + \Delta(b)$ are the two exactly tied, and (R3) breaks that measure-zero tie toward silence. It does so not to restrain her—she is indifferent, and loses nothing—but to keep the father's problem well posed: he will want to park as high in the band as he can, and were the tie broken the other way the parking payoff would be a supremum he could approach but never attain. The convention buys existence of the parking outcome, nothing more.

4.5 Separation, and the end of uncertainty

Separation must run through σ_1 (Lemma 5), and the question is how large an immediate sanction the strong type must impose for the weak not to follow. Define the *mimicry stake*

$$M(x) \;=\; \hat{\pi}\Big[(x-1)^2 - \Delta(b)^2\Big]\,:$$

as the proof below shows, $M(x)$ is exactly what the weak type gains, gross of that sanction, by passing as strong rather than standing revealed.

Lemma 8 (Separation)**.** *Let $b < a$ and $x > 1 + \Delta(b)$. A separating equilibrium surviving the Intuitive Criterion exists at every such reach. In the selected one the strong type plays $(\sigma_1^{\text{sep}}, \sigma_2^\star)$ with*

$$\sigma_1^{\text{sep}} \;=\; \frac{M(x)}{c_W},$$

the weak type plays $(0,0)$, a harsh father proceeds to the reach x against her, and is held at $1+b$ by a revealed strong mother.

Proof. Define the *value of credibility* at announced rate r: the harm a mother read as strong is spared, net of the posturing,

$$H_x(r) \;=\; \hat{\pi}\big[(x-1)^2 - (P_x(r)-1)^2\big] - \gamma r.$$

On the dead range $r \le 2\beta(\theta - x)$ the response is $P_x = x$ and $H_x(r) = -\gamma r \le 0$; on the saturated range $r > 2\beta a$ the pull is capped at $h = 0$ and H_x falls in r, dominated by the active endpoint. On the active range $r \in \big(2\beta(\theta-x),\, 2\beta a\big]$, write $h = P_x(r) - 1$, so $r = 2\beta(a-h)$ and $\gamma r = 2\hat{\pi}b(a-h)$; then $H_x = \hat{\pi}[(x-1)^2 - h^2 - 2b(a-h)]$, maximised at $h = b$, where

$$\max_r H_x(r) \;=\; \hat{\pi}\big[(x-1)^2 - (2ab - b^2)\big] \;=\; \hat{\pi}\big[(x-1)^2 - \Delta(b)^2\big] \;=\; M(x),$$

attained uniquely at $r = \sigma_2^\star$. So $M(x)$ is exactly the most that *being read as strong* can ever be worth, to either type, and the strong type's bundle attains it—the least-cost separating outcome in the sense of Riley (1979).

Three jobs remain, and it pays to keep them apart: incentive compatibility pins the immediate sanction from below; selection pins it from above and breaks a tie;

the refinement then only checks beliefs off the path.

Incentive compatibility: the sanction from below. Revealed, the weak type announces nothing—any rate is discounted to zero and the posturing is pure waste—for a loss of $\hat{\pi}(x-1)^2$. Mimicking a bundle (σ_1, r) read as strong costs her the posturing and the immediate sanction but never the execution, so her gain over revelation is $H_x(r) - c_W\sigma_1 \leq M(x) - c_W\sigma_1$. Any separating sanction paired with $\sigma_2^\star$ must therefore satisfy $c_W\sigma_1 \geq M(x)$: below that level she strictly mimics, which is why separation requires an immediate sanction at all.

Selection: the sanction from above, and the tie. At $\sigma_1 = M(x)/c_W$ the weak type is exactly indifferent between revealing and mimicking; revealing is then already a best response, and it is parsimony (R2), not the refinement, that selects it—the two replies leave her continuation payoff unchanged, the least signal is the non-signal, and the Intuitive Criterion disciplines beliefs after off-path deviations and has no purchase on an on-path indifference. Nor could the tie be broken the other way: a weak type playing the bundle turns the configuration into a pool, which Lemma 9 below removes. Immediate sanctions strictly above $M(x)/c_W$ also separate: the strong type does not bear them, and beliefs reading any lesser sanction as weak sustain them as perfect Bayesian equilibria surviving the refinement. They are, however, payoff-equivalent for the sender and strictly dearer for the father, and (R2) discards them for the least sufficient sanction.

Refinement: survival of the Intuitive Criterion. What remains for the refinement is the off-path beliefs at the selected configuration. A deviation (σ_1', r') is equilibrium-dominated for the weak type iff $H_x(r') - c_W\sigma_1' < 0$; for such deviations, when they are not also dominated for the strong type, the criterion requires the father to place no weight on the weak type. Where both types are ruled out the criterion is silent and beliefs are free; we read those deviations as strong as well. The strong type's gain from any deviation read as strong is $H_x(r') - M(x) \leq 0$: she already enjoys full credibility at the cheapest sufficient bundle, so no deviation the criterion forces to be read favourably strictly profits her. Deviations not dominated for the weak type may carry adverse beliefs, under which they are strictly unattractive to both types. The selected configuration survives. □

Remark 4. The separating sanction vanishes exactly at the edge of the reticence band: $\sigma_1^{\text{sep}} \downarrow 0$ as $x \downarrow 1 + \Delta(b)$, and $M(x) > 0$ precisely where the strong mother

deters at all. Separation and deterrence switch on together.

Lemma 9 (Uncertainty does not survive scrutiny)**.** *No pooling equilibrium survives the Intuitive Criterion at any reach $x > 1 + \Delta(b)$.*

Proof. Fix any candidate pool: both types post the same (σ_1^p, σ_2^p), the father retains $\hat{\rho} = \rho$ and responds with some level P^p. Since neither type bears an execution cost and the immediate sanction enters separably, the part of the pooled loss *net of that sanction* is identical across types,

$$\Lambda^p \;=\; \hat{\pi}\,(P^p - 1)^2 + \gamma\sigma_2^p,$$

the weak type paying $c_W\sigma_1^p$ on top. Compare Λ^p with the corresponding net loss of full credibility at the cheapest sufficient rate, $\hat{\pi}\Delta(b)^2$. Writing $b_p = \beta\gamma/(\hat{\pi}\rho) > b$, the best net loss *any* pool can deliver is $\hat{\pi}\min\{(x-1)^2,\, \Delta(b_p)^2\}$ when $b_p < a$—deterring at diluted credibility forces an inflated announcement for the same pull, and whether it beats idling depends on the reach—and simply $\hat{\pi}(x-1)^2$, idling, when $b_p \geq a$. Since Δ is strictly increasing and $x > 1 + \Delta(b)$, both candidates strictly exceed $\hat{\pi}\Delta(b)^2$, so in either case

$$G^p \;\equiv\; \Lambda^p - \hat{\pi}\Delta(b)^2 \;>\; 0:$$

both types would gain exactly G^p, gross of any change in the immediate sanction, from being read as strong at the bundle $(\sigma_1', \sigma_2^\star)$.

Now use that σ_1 is unbounded: choose $\sigma_1' > \sigma_1^p + G^p/c_W$. For the weak type the deviation's value, even under the most favourable belief, is $G^p - c_W(\sigma_1' - \sigma_1^p) < 0$: it is strictly equilibrium-dominated. For the strong type the sanction is free, so under the belief $\hat{\rho} = 1$ the deviation yields exactly $G^p > 0$. The Intuitive Criterion therefore requires the father to read the deviation as strong, the strong type strictly profits, and the pool fails. The argument covers the silent pool $(0, 0)$, where $\Lambda^p = \hat{\pi}(x-1)^2$ and $G^p = M(x) > 0$. □

Remark 5 (What carries the no-pooling result). The lemma leans on two structural features, and both should be read in the open: the immediate sanction σ_1 is unbounded, and it is free to the strong type, $c_S = 0$. With a bounded sanction $\sigma_1 \leq \bar{\sigma}_1$, the pool-breaking deviation exists only if $\bar{\sigma}_1 - \sigma_1^p > G^p/c_W$; where the bound bites, pools can survive, and with them the weak mother's shelter. With a

small positive cost $0 < c_S < c_W$ for the strong type—bracketing its knock-on effect at the final stage, Remark 2—the deviation requires a sanction increment d with $G^p/c_W < d < G^p/c_S$, which exists—in this static comparison; the fully perturbed game also moves execution and pooling payoffs and is not solved here—precisely because $c_S < c_W$. The engine is thus not unboundedness alone but an unbounded signal with a strict cost advantage for the resolute type; $c_S = 0$ is its cleanest case, in which proof is always technologically available and costly only for the wrong type.

Remark 6 (The costless-talk limit). The lemma's engine is a strict gain, and it is bought by $\gamma > 0$. At $\gamma = 0$ a pool on a sufficiently high rate deters fully at zero cost to both types, nothing can strictly improve on it, and fully protective pooling survives the refinement. The selected equilibrium does not converge to that outcome as $\gamma \to 0$: the band vanishes, but the weak mother's son's punishment tends to $x^\dagger$, which is free of γ. The model's tragedies are bought by an arbitrarily small posturing cost.

4.6 The continuation at a given reach

What this section has assembled can now be stated as one object—the continuation the father's first strike will price.

Proposition 1 (The continuation at a given reach)**.** *Fix $\hat{\pi} > 0$ and a reach $x > 1$, and write $b = \beta\gamma/\hat{\pi}$; at $\hat{\pi} = 0$ the mothers' objective carries no harm term and parsimony selects silence for both types. If $b \geq a$, the selected continuation is* quiet *at every reach: no sanction from any mother, a harsh father takes the reach. If $b < a$, the selected surviving continuation is unique: quiet for $x \leq 1 + \Delta(b)$, and* separation *for $x > 1 + \Delta(b)$, as in Lemma 8. In either case the weak mother's son is never sheltered: where resolve can always be proven, irresolution can always be detected.*

Proof. If $b \geq a$, deterrence pays at no reach and no credibility: even at full credibility the active-range objective is increasing (Lemma 7), being read as strong is worth nothing, and by parsimony no mother sanctions or announces; the father takes the reach. Let then $b < a$. Below the band no deterrence is worth its posturing at any credibility (Lemma 7, with Δ increasing), so by parsimony all are silent. Above it, Lemma 9 removes every pool and Lemma 8 provides the separating equilibrium. Separation at any other announcement also fails: a

strong type held to a credibility gain $K < M(x)$ can deviate to $\sigma_2^\star$ paired with an immediate sanction large enough that the weak type's gain $M(x) - c_W\sigma_1'$ turns negative; the criterion then reads the deviation as strong and she collects $M(x) - K > 0$. The announcement is thus pinned at $\sigma_2^\star$, and incentive compatibility with parsimony pins σ_1^{sep}; by Lemma 6 the weak mother then helps her son not at all. □

5 The father's first strike

We now step back to the father's opening move, taking Proposition 1 as the continuation. Because $\bar{p} < 1$ and $\bar{p} < \theta$, neither ideal can be delivered in one round: the first strike commits the father's reach $x = p_1 + \bar{p}$.

The just father. The just father attains his ideal at every opening whose reach lies in $[1, 2\bar{p}]$: there he can set $P = 1$, and by Lemma 2 he does. He does not open below reach 1, where his reach would fall short of the just level and he would undershoot $P = 1$ at a strict loss; and by parsimony (R2) he does not arm above what his ideal requires. He therefore opens at the least sufficient reach, reach 1. This is not innocuous, and the skeptical question deserves an answer in the open: could he not instead arm higher, pooling with the harsh type at some quiet-band reach and leaving the mother uncertain whom she faces? He could, at no cost to himself—but parsimony selects his least opening, and the choice is substantive: it makes the first move maximally informative. Opening at reach 1 vacates every reach above the just level for the harsh type alone, which is what licenses the belief $\hat{\pi}(x) = 1$ of (R3); had the just father pooled higher, the posterior $\hat{\pi} < 1$ would inflate $b = \beta\gamma/\hat{\pi}$ and, at any fixed reach, widen the band of unanswered aggression. That is a fixed-reach comparison, and it is all the selection buys: once the harsh father chooses his reach against the continuation, the welfare effect of the least opening is ambiguous—there are pooled openings, excluded here, under which he settles for less than the unmasking would hand him. No sanction is ever levied at his opening, the total being capped at 1 regardless of type (Lemma 4). For the remainder,

$$\hat{\pi} = 1, \qquad b = \beta\gamma, \qquad \Delta = \Delta(b), \qquad M(x) = (x-1)^2 - \Delta^2,$$

with $b < a$ by Assumption 2.

The value of a reach. The harsh father chooses $x \in [1, \theta]$, every value feasible by Assumption 1. His expected loss is

$$V(x) = \begin{cases} \beta(\theta - x)^2, & x \le 1 + \Delta, \\ L(x) \;=\; (1-\rho)\,\beta(\theta - x)^2 + \dfrac{\rho}{c_W}\,M(x) + \rho\,\beta\big(a^2 - b^2\big), & x > 1 + \Delta, \end{cases}$$

the separating branch collecting, as before, the immediate sanction $M(x)/c_W$ he absorbs, the quadratic loss at the hold level, and the executed sanction: $\beta(a - b)^2 + \sigma_2^\star b = \beta(a^2 - b^2)$.

Lemma 10. (i) *On the quiet band V is strictly decreasing.*

(ii) *Entering separation is discretely costly: at $x \downarrow 1 + \Delta$, L exceeds the quiet value by $2\rho\beta a(\Delta - b) > 0$; the lower corner of the separating range is never optimal.*

(iii) *L is strictly convex with unconstrained minimiser*

$$x^\dagger = \frac{(1-\rho)\,\beta c_W\,\theta + \rho}{(1-\rho)\,\beta c_W + \rho} \;<\; \theta, \qquad x^\dagger - 1 = \frac{(1-\rho)\beta c_W}{(1-\rho)\beta c_W + \rho}\,(\theta - 1).$$

Proof. (i) Immediate. (ii) At $x = 1 + \Delta$ the separating sanction vanishes (Remark 4) and the gap is $\rho\beta[(a^2 - b^2) - (a - \Delta)^2] = 2\rho\beta a(\Delta - b)$, using $\Delta^2 = 2ab - b^2$. (iii) The first-order condition $-2(1-\rho)\beta(\theta - x) + 2(\rho/c_W)(x - 1) = 0$; the Δ^2 and constant terms drop out. □

The father's problem is thus a contest between exactly two candidates: parking at the top of the quiet band, $x_A = 1 + \Delta$, or striking at the interior optimum $x^\dagger$. Two conditions, both explicit in primitives, decide it. Write $m = (1-\rho)\beta$, $q = \rho/c_W$, and

$$\text{(P1)} \qquad (1-\rho)\,\beta c_W \;>\; \frac{\rho\,\Delta}{a - \Delta}, \tag{1}$$

$$\text{(P2)} \qquad \frac{mq}{m+q}\,a^2 \;-\; q\,\Delta^2 \;+\; \rho\beta\big(a^2 - b^2\big) \;<\; \beta\,(a - \Delta)^2. \tag{2}$$

(P1) is $x^\dagger > 1 + \Delta$: the interior strike exists. (P2) is $L(x^\dagger) < V(x_A)$: striking beats parking, the left side being the closed form of $L(x^\dagger)$, its first term the harmonic-weighted minimum of the two quadratics.

Neither condition is free of γ, though the notation may hide it: $\Delta = \Delta(b)$ with

$b = \beta\gamma$, so both (P1) and (P2) move with the posturing cost. As γ falls, Δ falls with it; the right side of (P1) collapses toward zero and (P2) relaxes alongside—which is what powers the small-γ claim of the theorem below. Read in primitives, the unmasking regime is the land of cheap threats and dear follow-through: posturing inexpensive, so the band is thin and parking buys little; resolve rare and the weak mother's execution cost heavy, so the composite $(1-\rho)\beta c_W$ carries (P1). Where threatening is dear, the band swallows everything; where follow-through costs no one much, there is little to sort and the interior strike loses its edge.

Theorem 1 (The harsh father's strike, and the son's fate)**.** *Under Assumptions 1–2, in the selected pure-strategy perfect Bayesian equilibrium of Definition 1, for any $\gamma > 0$ the harsh father strictly arms: his reach satisfies $x^* \geq 1 + \Delta > 1$, so disarming—mimicking the just father—is never played. Moreover:*

(i) The unmasking (separation). *If* (P1) *and* (P2) *hold, he strikes $x^* = x^\dagger$; the mothers separate; the son is held at $1 + \beta\gamma$ when his mother is strong and driven to*

$$P_{\text{weak}} \;=\; x^\dagger \;=\; \theta - \frac{\rho\,(\theta - 1)}{(1-\rho)\,\beta c_W + \rho}$$

when she is weak.

(ii) The band (silence). *Otherwise he parks, $x^* = 1 + \Delta$; no sanction is ever levied, and the son's punishment is $1 + \Delta(\beta\gamma)$ under* both *mother types.*

Both conditions (P1) *and* (P2) *hold on an open set of primitives; each depends on γ through $\Delta = \Delta(\beta\gamma)$, and for any β, ρ, c_W they hold for all γ small enough.*

Proof. Reach 1 is the left endpoint of the quiet band, on which V is strictly decreasing and sanction-free, and $\Delta > 0$: he never disarms. By Lemma 10 the optimum is x_A or the projection of $x^\dagger$; the projection is interior iff (P1), in which case (P2), holding strictly, decides; at equality in (P2) both reaches are optimal and (R3) selects parking. When (P1) fails the corner is dominated by x_A via the jump. Outcomes are Proposition 1 and Lemma 8. Non-vacuousness: as $\gamma \to 0$, $b \to 0$ and $\Delta \to 0$; (P1) tends to $(1-\rho)\beta c_W > 0$, and (P2) tends to $\frac{mq}{m+q}a^2 + \rho\beta a^2 < \beta a^2$, which, dividing by a^2 and using $m = (1-\rho)\beta$, is $\frac{mq}{m+q} < m$, i.e. $\frac{q}{m+q} < 1$—both strict. □

Remark 7 (Sanity check at the extremes). As $\beta\gamma \to 0$ and $(1-\rho)\beta c_W/\rho \to \infty$, along any path on which the unmasking remains selected, the fates sort completely: the strong mother's son is held within any ε of the just level, the weak mother's son driven within ε of θ. In the depth of the unmasking regime, deterrence does not fail on average—it *sorts*: the son's fate is decided entirely by his mother's resolve, and the father arms hard precisely because weakness will be found out.

Proposition 2 (Comparative statics)**.** *In the unmasking regime:*

(i) $P_{\text{weak}} = x^\dagger$ *is strictly increasing in* β, c_W *and* θ, *strictly decreasing in* ρ, *and independent of* γ;

(ii) *the strong-branch outcome* $1+\beta\gamma$ *is increasing in* β *and* γ;

(iii) *the expected punishment level* $\rho(1+\beta\gamma)+(1-\rho)\,x^\dagger$ *is strictly decreasing in* ρ *and strictly increasing in* β.

In the band regime, the son's punishment $1+\Delta(\beta\gamma)$ *is strictly increasing in* β *and* γ *under both mother types.*

Proof. (i) Write $A=(1-\rho)\beta c_W$, so $x^\dagger - 1 = a\,A/(A+\rho)$, increasing in A and in a. Explicitly,

$$\frac{\partial x^\dagger}{\partial \rho} = -\frac{a\,\beta c_W}{\big((1-\rho)\beta c_W+\rho\big)^2} < 0, \qquad \frac{\partial x^\dagger}{\partial \beta} = \frac{a\,(1-\rho)c_W\,\rho}{\big((1-\rho)\beta c_W+\rho\big)^2} > 0,$$

and symmetrically for c_W; monotonicity in θ is monotonicity in a. (ii) Immediate. (iii) In ρ the derivative is $(1+\beta\gamma)-x^\dagger+(1-\rho)\,\partial x^\dagger/\partial\rho$, and both pieces are negative: the first because $x^\dagger > 1+\Delta > 1+b$ in the regime, the second by (i). In β both branches rise: the strong branch at rate $\rho\gamma>0$, the weak branch at rate $(1-\rho)\,\partial x^\dagger/\partial\beta > 0$. The band statement is $\Delta'(z) = (a-z)/\Delta(z) > 0$. □

These are within-regime statements, and the caveat matters most for γ. The independence of $x^\dagger$ from γ holds only while the unmasking is selected: raising γ leaves $x^\dagger$ in place but lifts the band top $1+\Delta(\beta\gamma)$ toward it, and the equilibrium switches into the band—(P2) fails strictly before (P1) can bind—where γ is anything but neutral. The handover is taken up in Section 6.

One derivative in (i) deserves to be lifted from the list, for it is the least intuitive

in the paper: P_{weak} *rises* in c_W. One might have guessed the opposite—the weak mother's burden, surely, is her own affair. It is not, and the chain runs through the father's pocket. A larger c_W makes mimicry dearer, which cheapens the strong mother's proof: the separating sanction $\sigma_1^{\text{sep}} = M(x)/c_W$ falls. That sanction is precisely what the father absorbs when the mother proves strong, and as it shrinks, so does his reason for restraint: he arms harder, and the reach he commits—the reach the *weak* mother's son will absorb in full—grows. The parameter that makes resolve provable is the one that drives the unprotected son's exposure; Section 7 gives the phenomenon its name—loosely, the father's information rent, collected from the boy.

Corollary 2. *Any path that offers the son any protection—any outcome in which the father is held short of his committed reach—necessitates a sanction in period one.*

Proof. By Proposition 1 the only continuation in which the father is held short of his reach is separation, which requires the positive immediate sanction $\sigma_1^{\text{sep}} = M(x)/c_W > 0$ (Lemma 8); the announced rate alone cannot separate (Lemma 5) and no pooled announcement survives (Lemma 9). Note the timing: $p_1 = x - \bar{p} \leq \bar{p} < 1$ by Assumption 1, so the sanction is levied while the punishment inflicted still lies below the just level. □

The corollary speaks of holding the father short of his chosen reach. The reach itself is already trimmed by the sanction he expects to absorb (Proposition 2: $x^\dagger$ falls in ρ): the early sanction protects once when feared and once when paid, and the weak mother's son knows only the first kind.

Remark 8 (One-sided equivalence). The reader may ask what the father's privacy contributes, since he surrenders it with his first move. In terms of outcomes: nothing. Set $\pi = 1$—a father known to be harsh from the outset—and the selected equilibrium is unchanged: the continuation after any reach already runs at $\hat{\pi} = 1$, and the father's first-strike problem is independent of π. What the two-sided model carries is not an outcome but a moment—the moment at which certainty is acquired. At $\pi < 1$, the mother's certainty that excess is coming is produced by the father's own opening: a strike still below the just level which, read against the just father's least opening, proves the type, and with it the intention. Two things hang on this moment. The first is normative and taken up in the coda: the early sanction of Corollary 2 answers a proven intention, not a suspicion. The second

is strategic: the target excess is $b = \beta\gamma/\hat{\pi}$, so the band of tolerated aggression is widest while the father might yet be innocent and narrowest at $\hat{\pi} = 1$. Evidence is a deterrence asset, and the father hands it over himself. Both rest on the selection: it is the just father's least opening that vacates every higher reach for the harsh type (Remark 1); where innocents may over-arm, proof decays into evidence.

6 Zeal

Zeal is the model's most malign parameter, and its subtlest. The malignity is within-regime and quickly recorded; the subtlety lives at the boundary between the regimes, where it reverses a sign.

Corollary 3 (Zeal)**.** *The father's intensity is malign through every channel that exists. Within the unmasking regime it raises both branches at once; within the band regime it raises the unanswered reach; and in the joint limit $\beta\gamma \to \theta - 1$ the son's punishment converges to the harsh ideal θ under* both *mother types and in* whichever *regime is selected.*

Proof. The first two claims are Proposition 2. For the limit, the selected outcome is $1 + \Delta$ in the band and $\{1 + b,\ x^\dagger\}$ in the unmasking, where (P1) places $x^\dagger$ strictly *above* the band top: $1 + \Delta < x^\dagger \leq \theta$ whenever the unmasking is played. As $\beta\gamma \to a$, both $1 + b \to \theta$ and $1 + \Delta \to \theta$, and the sandwich $1 + \Delta < x^\dagger \leq \theta$ forces $x^\dagger \to \theta$ along any sequence on which the unmasking is selected. Which regime survives near the limit depends on the path: $x^\dagger$ is γ-free, so raising γ at fixed (β, ρ, c_W) leaves it in place while the band top rises toward it, and (P2) fails before (P1) can bind: where (P1) binds, $x^\dagger = 1 + \Delta$, and the entry jump of Lemma 10(ii) gives $L(x^\dagger) = V(x_A) + 2\rho\beta a(\Delta - b) > V(x_A)$, so (P2) has already failed—by continuity, strictly before. The band therefore takes over before the limit is reached, and the convergence claim is indifferent to the handover. □

Within each regime, then, more zeal is more harm. Across the handover it is not, and the reason is an accounting identity worth stating as a result.

Proposition 3 (The handover)**.** *Consider any primitives at which* (P1) *holds and* (P2) *sits exactly on its boundary, $L(x^\dagger) = V(x_A)$, so that the father is indifferent between parking and the separating strike. The son's expected excess under the strike strictly*

exceeds his excess under parking:

$$\rho\, b + (1-\rho)\,(x^{\dagger} - 1) \;>\; \Delta.$$

Proof. Write $h = P - 1$ for the son's excess. Under parking $h = \Delta$ surely and no sanction is levied, so the father's loss is $\beta(a-\Delta)^2$. Under the strike his loss carries, besides the punishment terms, the sanctions he absorbs, $\mathbb{E}[S] = \rho\left[\sigma_1^{\text{sep}} + \sigma_2^{\star} b\right] > 0$. Indifference therefore gives $\beta\,\mathbb{E}\left[(a-h)^2\right] = \beta(a-\Delta)^2 - \mathbb{E}[S] < \beta(a-\Delta)^2$. By Jensen's inequality $(a - \mathbb{E}[h])^2 \le \mathbb{E}[(a-h)^2]$; and since the father never strikes past his ideal, $h \le a$, both $a - \mathbb{E}[h]$ and $a - \Delta$ are non-negative, so taking roots preserves the ranking: $a - \mathbb{E}[h] < a - \Delta$, that is $\mathbb{E}[h] > \Delta$. Under the strike $\mathbb{E}[h] = \rho b + (1-\rho)(x^{\dagger} - 1)$. □

Corollary 4. *Wherever a path through the primitives crosses the handover, the son's expected punishment jumps with it, and the unmasking side of the boundary is the darker one. In particular, where rising zeal ends the gamble, the son's expected punishment falls.*

And zeal does end the gamble. It also begins it: at small β the interior strike does not exist, since the left side of (P1) scales as β while the right side, through $\Delta = \Delta(\beta\gamma)$, scales as $\sqrt{\beta}$; at large β the corner argument in the proof of Corollary 3 shows that (P2) fails strictly before (P1) can bind. The unmasking, where it exists at all, is a mid-zeal phenomenon: too little zeal and the prize is not worth its toll; too much and the band's rising edge hands the father his fill without a fight. Screening is the calculating father's regime, not the zealot's. Nothing here claims the screening region is a single interval; the two asymptotics say only that it needs intermediate zeal. Its lower boundary can sit very near zero—at the parameters of Figure 2 it is invisible—but it is there, and it moves where the weak mother's cost is small.

Figure 2 draws the path. Within each regime the curve only rises—zeal is malign on both branches, and in the limit the band's excess reaches the harsh ideal. The one descent is the handover: where the gamble ceases to pay, the father parks, and the son's expected punishment falls. The fall is an un-sorting. The strong mother's son loses his shelter; the weak mother's son escapes the deep reach; and the average tilts kind at every admissible mix of mothers, because the father's indifference is priced in sanctions the son never sees.

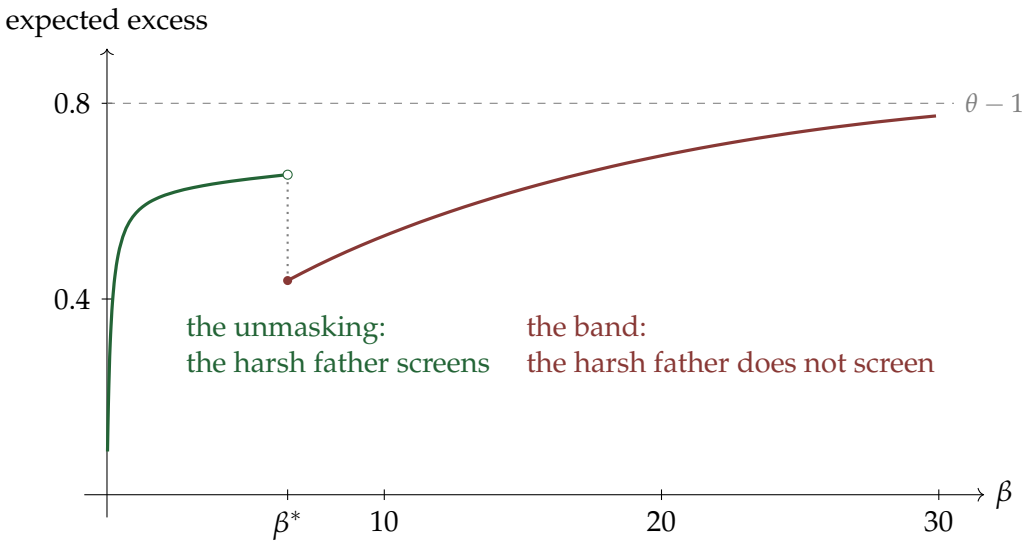


Figure 2: The son's expected excess, conditional on a harsh father, as the father's zeal grows; drawn at $\theta = 1.8$, $\bar{p} = 0.95$, $\rho = 0.2$, $\gamma = 0.02$, $c_W = 2$. Within each regime harm rises; at the handover $\beta^* \approx 6.5$, where the gamble ceases to pay, it falls discretely (Proposition 3). As $\beta\gamma \to \theta - 1$ the band's excess approaches the harsh ideal.

7 Two ways to tragedy

Theorem 1 leaves the harsh father exactly two configurations, and they are the two ways to tragedy.

The band: quiet harm. In the first (trivial) way to tragedy the harsh father parks at the top of the reticence band, $1 + \Delta(\beta\gamma)$, and *the mothers stay silent*: no immediate sanction, no announced sanctioning scheme, leaving the son—whoever his mother is—exposed to a punishment exceeding the just level. The mother is silent not because she is weak but because answering is not worth her posturing; the father, knowing the exact boundary of what will be tolerated, stands on it. As threatening grows dearer or the father more zealous, the band's edge climbs, and at $\beta\gamma \to \theta - 1$ the harsh father can reach his ideal.

The unmasking: sorted harm. In the second (less trivial) way to tragedy the harsh father exposes himself, screening the mothers and unmasking the weak one. The resolute mother proves herself and holds the son at $1 + \beta\gamma$; the weak mother's son, by contrast, absorbs the father's entire chosen reach. Fates sort, and the sorting is the tragedy: the son's punishment is decided not by what he did, nor even by the father alone, but by whether his mother can afford to be believed.

What restrains the father, and what does not. The harsh father's restraint is priced by a single object—the sanction $M(x)/c_W$ the father absorbs when the mother proves strong.

Three readings of the same formula $x^\dagger = \theta - \rho(\theta - 1)/[(1-\rho)\beta c_W + \rho]$ organise the comparative statics.

Credibility restrains before the fact: a larger ρ lowers the damage *even conditional on the mother proving weak*, through the reach the father declined while he still fears the presence of the strong mother.

Zeal dissolves restraint: as β grows the father ceases to weigh the sanction, and $x^\dagger \to \theta$ as $(1-\rho)\beta c_W$ grows without bound. And the weak mother's pain is the father's information rent: a larger c_W cheapens the strong mother's proof, shrinks the sanction the father absorbs, and so *emboldens* him—the very parameter that makes resolve provable is the one that drives the unprotected son's exposure.

What resolve must be made of. The assumption $c_S = 0$ may look like a normalisation. It is not. The cleanest way to see what it carries is to solve the model it excludes.

Suppose instead that sanctions cost both mothers alike, $c_S = c_W = c > 0$, and let the strong mother be a commitment type in the tradition of Kreps and Wilson (1982): an automaton who executes whatever she has announced. Resolve is then perfectly real, but it shows itself only at the moment of execution—and that is too late for the machinery. The immediate sanction now costs both types the same, so it can no longer separate them; the argument of Lemma 5 extends to it verbatim.

The announcement channel fares worse: it runs backwards. The committed type expects to pay $c\,\sigma_2(P-1)$ when she executes; the weak type knows she will renege and bears only the posturing. Announcing is now *dearer for the resolute type*—the wrong Spence ordering, with mimicry cheap for exactly the type one wishes to screen out. And the deviation that broke the pools in Section 4 required, as Remark 5 recorded, an unbounded signal with a strict cost advantage for the resolute type. Here there is none. The fully deterring pools stand, and standing, they can protect: where the inflated rate is worth its posturing, it is paid for in posturing alone and never executed—a deterred father leaves nothing to execute on.

So that world loses the sorting, not the protection. Nothing in it separates; no mother is ever found out, and none is ever proven. Resolve that shows itself only at execution protects, if at all, through uncertainty—the very uncertainty that scrutiny destroys in our model. $c_S = 0$ is the cleanest world in which proof exists; it is also the world in which no one can hide.

The price of proof. The commitment variant changes several things at once; a cleaner comparison removes only the instrument. Let the immediate sanction be unavailable and keep every payoff as it is. Separation then loses its engine: announcing costs both types the same, so no mother can strictly prefer to reveal herself. The knife edges recorded in Lemma 5 do survive—separations held together by indifference and adverse beliefs—but the selection below passes them over. The pools that scrutiny destroyed in Section 4 stand.

Survival brings multiplicity. A silent pool, in which no mother speaks and any announcement is read as weakness, is an equilibrium too, and there the harsh father simply takes θ. The comparison therefore needs a selection of its own, and we give the mothers the analogue of what the father's world had: among the surviving pools, the one they themselves prefer. On this the two types agree—the father discounts every announced rate by ρ, so both want the pooled rate that pulls him to the inflated target $b_p = \beta\gamma/\rho$.

The father's opening is then immediate. Being held at $1 + b_p$ while absorbing the expected execution costs him $\beta(a^2 - b_p^2)$; parking at the pooled band's edge costs $\beta(a - \Delta(b_p))^2$; and parking wins whenever $b_p < \Delta(b_p)$, which is always. So under this benchmark the harsh father parks. Every son receives $1 + \Delta(\beta\gamma/\rho)$, and no sanction is ever paid. (When $\beta\gamma \geq \rho a$ even the pooled threat is not worth its posturing, and he simply takes θ.)

The comparison with Theorem 1 is the point. Making resolve provable does two things at once: it cheapens deterrence, narrowing the quiet region from $\Delta(\beta\gamma/\rho)$ to $\Delta(\beta\gamma)$, and it hands the father the screen. Where he parks, proof helps every son, strictly. Where he screens, it can hurt them on average—at the parameters of Figure 2 with $\beta = 1$, expected punishment is 1.39 without the instrument and 1.57 with it, and the weak mother's son bears the difference. Against the silent pool the ranking reverses: there the father takes $\theta = 1.8$, and proof lowers expected punishment to 1.57.

The ranking is a fact about the selected equilibrium, not about proof as such. The over-sized burns that (R2) discards would tip the father into parking, at no mother's expense—the strong type carries them for free, the weak type never carries them at all—and every son would then stand at $1 + \Delta(\beta\gamma)$, below anything the world without the instrument offers. What the comparison condemns is thus not proof itself but the parsimony with which the selection buys it.

What proof is worth thus depends on what uncertainty would otherwise have done: against protective uncertainty it can harm, against organised silence it helps. Protective uncertainty was never the enemy; it was a shelter, and proof is the technology that tears it down.

Protection must be paid for early. Every path on which the son is protected runs through the period-one sanction (Corollary 2). And the sanction comes early by necessity: it is paid while the punishment inflicted so far still lies below the just level, answering not an excess—none has yet occurred—but a committed reach. This is the model's most uncomfortable political lesson. Whoever wants to protect must be ready to sanction before the wrong, on the evidence of preparation alone; a third party that waits until the excess has materialised has nothing left to prove and no one left to deter. Even the father's restraint short of his ideal is priced by the same early instrument: what holds his reach down is the sanction he expects to absorb should the mother prove strong. One bracket belongs on the lesson: the two-period calendar offers no intervention date after a first excess, so the model cannot distinguish sanctioning before the wrong from sanctioning without delay; the prescription attaches to the stated game.

What the model finally says. Conditional on a harsh father, the son is never punished at the just level, and near it only when the punishment motive itself is faint or when his mother is resolute and the father's strike gives her the occasion to prove it. Every other path is one of the two tragedies: the band, where harm is quiet and universal; the unmasking, where the weak mother's son gets exposed and suffers badly. Both ways to tragedy deepen in the father's zeal, each within its regime; each also deepens in its own cost—the band in the price of threatening, the unmasking in the weak mother's price of acting. The regions of greatest harm are intuitive: a zealous father, expensive threats, rare resolve, and a pretence too painful to maintain—there the model places the weak mother's boy at the edge of θ.

8 Coda

The formalism is done; what follows is interpretation, and the reader who wants only the theory may stop here.

Gaza. The model was written with Gaza in view, and its mapping is as plain as it is partial. A power inflicts punishment on a population beyond what even sympathetic observers would call just; states (or communities of states) weigh sanctions. Some announce threats but only those who sanction early are believed. In the case of Gaza early sanctions did not materialise and excessive harm unfolded—with over 70,000 Palestinians being killed, among them more than 20,000 children.

Red lines, and due process. A reader of an earlier draft pointed the model at another episode: the American red line over chemical weapons in Syria—an announcement, prominent and public, with no sunk cost attached, and in the end no execution. From the vantage point of this paper's model this hardly comes as a surprise. Where an initial sanction would have been on the table, talk about future sanctions is not merely cheap—it is testimony against the speaker. That reading is the selected equilibrium's: its beliefs treat an announcement unaccompanied by a burn as the weak type's instrument.

Yet the model's prescription—sanction before a verified escalation to excessive harm, on the evidence of some inflicted harm alone—collides with norms of due process. The collision is softer though than it appears, and the model says why. By the time the mother sanctions, she is not acting on mere suspicion: the father's opening, read against a just father who strikes no harder than his purpose requires, has proven his type, and with it the excess to come. The harsh father's punishment in round 1 lies below the just level but his intention is revealed without doubt. Whether such an inference can be encoded in actionable law is a question I defer to legal scholars.

Beyond two periods. A reader may suspect that the immediacy of the father's self-revelation is an artifact of the calendar: with only two rounds, imitating the just father means opening at his least reach, and reach 1 is terminal—there is no later round in which to strike, so hiding costs the harsh father everything he wants. Grant more rounds, with the per-round capacity scaled down, and hiding becomes affordable: the harsh father can walk the just father's path for a while, and the equilibrium path may well contain such a prelude of pooling. But walking

it is not free. Every round spent in the corridor erodes the reach—delay disarms—and the prelude must end: a round will come, resembling our first, in which the harsh father shows his face while the punishment inflicted still lies below the just level, and from that round onward the harsh father faces, at whatever reach he has kept, a game of the kind solved above.

The weak mother. One sanctioner sits uneasily close to the model's weak type. Germany has declared Israel's security its *Staatsräson*, and the declaration has a candidate translation in the model's parameters: it is c_W—the cost of sanctioning at all, made immense by history; what it prices in equilibrium is not just a withheld execution but, more importantly, the early signal she cannot afford. The model does not say such a mother is insincere; her concern may be entirely earnest. It says she will be *found out*—and in any equilibrium where she is screened, she will leave her son completely unsheltered.

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